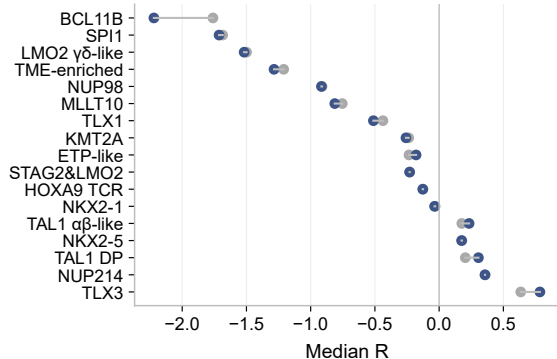


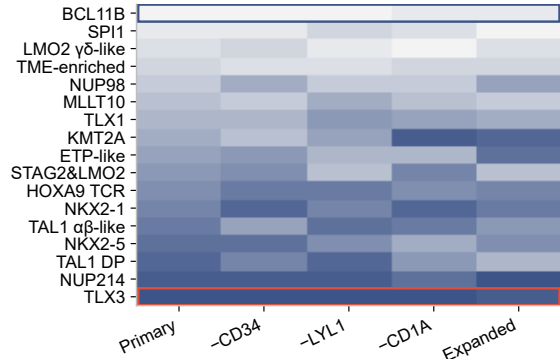
A Leave-subtype-out comparison

Shared baseline versus subtype-held-out spline



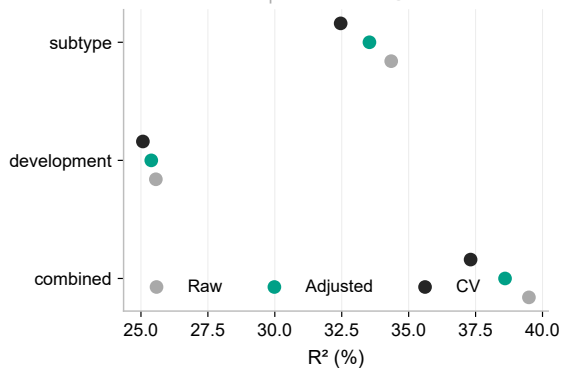
B Seventeen-subtype rank heatmap

Rank 1 = most ZEB2-skewed; middle ranks may move



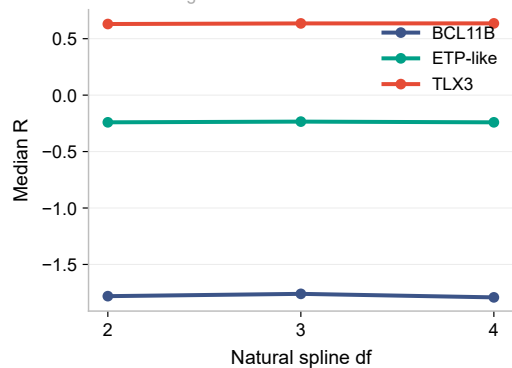
C Model-complexity audit

Patient-level repeated five-fold CV



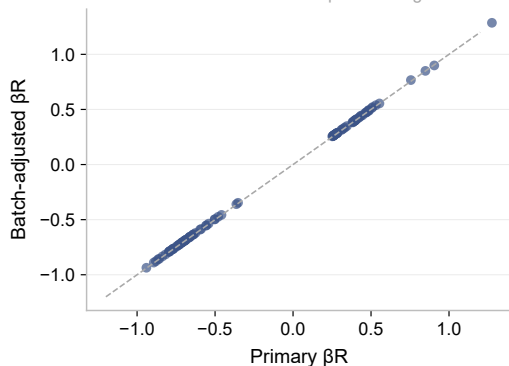
D Prespecified spline flexibility

No setting selected for favorable result



E Batch effect audit

100/100 frozen coefficients preserve sign



F Normal-reference domain

20 units are not 20 donors; 14/18 BCL11B outside

